# Abbott Unified Hypothesis (AUH): Experimental Testing Protocol & Verification Guide

This document serves as the official testing and measurement protocol for the empirical stress tests and predictions defined in the v23 Master Specification. These protocols are designed to ensure high-fidelity replication by focusing on raw substrate mechanics, direct calorimetry, and hardware-level displacement limits rather than probabilistic approximations.

## 1. The Tau-Proton Depth Challenge (The Causal Floor)

- **Objective:** Verify the compressible substrate gradient and the absolute compression floor of the proton.
- **Methodology:** High-precision tauonic-hydrogen scattering experiments.
- **AUH Prediction:** The effective proton radius under maximum tau-lepton mass load will compress to **0.8068 fm**, derived via the Causal Limit (Mc = 21,313.79 MeV).
- **Protocol Note:** Because the tau lepton has an extremely short lifespan (roughly 10^-13 seconds), transient scattering cross-section analysis must capture the collision depth before structural rupture occurs. Optical high-frequency probes (such as deep-UV lasers) only capture the first gear shift (0.84 fm Mechanical Core) and cannot reach this causal floor.

## 2. The Ghost Mirror Parity Challenge (Bismuth vs. Tellurium Calorimetry)

- **Objective:** Prove that radioactive decay is a deterministic pressure bleed rather than a probabilistic random event.
- **Methodology:** High-sensitivity cryogenic bolometer measurements.
- **Protocol:** Prepare exactly **1 mole of Bismuth-209** and **1 mole of Tellurium-130**. Isolate both samples in calibrated cryogenic bolometers.
- **AUH Prediction:** Despite possessing vastly different half-lives (10^19 years vs. 10^24 years), the bolometers will record **identical Total Thermal Energy Flux (X)** per second. Bismuth's slow, high-energy Alpha fractures mirror Tellurium's rapid, low-energy Beta seepage to maintain discharge parity against the Lead-82 static anchor.

## 3. The Helium-4 Hardware Reset (15.01% Volumetric Redline)

- **Objective:** Detect non-linear harmonic yield points under extreme pressure loading.
- **Methodology:** High-pressure laser spectroscopy of Helium-4 nuclei.
- **AUH Prediction:** As pressure approaches the **15.01% Volumetric Redline**, the Helium-4 radius will not shift along a smooth, linear curve. Instead, it will reach its structural yield point and trigger a discrete, non-linear **harmonic jump** (a sudden reset of the proton-neutron skin).

## 4. The 13 TeV Collider Parity Prediction (The Hard-Floor Metric)

- **Objective:** Audit particle creation as a deterministic integer division of substrate tension rather than a probabilistic cross-section.
- **Methodology:** High-energy proton-proton collisions at the Large Hadron Collider (LHC).
- **AUH Prediction:** In a **13 TeV (13,000,000 MeV)** impact, the substrate will deterministically generate exactly **304 stable proton-antiproton pairs** per direct impact, governed by the 42,627.58 MeV Structural Parity Threshold (E / 42,627.58). Remaining energy is vented as deterministic kinetic exhaust (meson shrapnel).

## 5. The SPT2349-56 "Scorching Cloud" Verification

- **Objective:** Account for thermal anomalies in infant cosmic structures without invoking unmapped dark matter forces.
- **Methodology:** Spectral analysis of intra-cluster gas at high redshift ($z > 4$).
- **AUH Prediction:** The observed 500% temperature excess will correlate exactly with the **0.1280 SI Tension Drop** as cosmic gas transitions from the 0.88 fm relaxed atmosphere to the 0.922 fm Abyss Gear Ratio during the universal piston cycle.

## Universal Constants Reference for Testing

- **Substrate Baseline:** 171.09 MeV/fm^3 (Maximum yield point of stable nuclear matter)
- **Causal Limit (Mc):** 21,313.79 MeV (Absolute tension ceiling before medium rupture)
- **Volumetric Redline:** 15.01% (0.1501 maximum expansion limit for a stable stall)
- **Lead-208 Anchor:** 1.000 SI (The universal zero-point static lock at 18.26 cm^3/mol)